John Pelan

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https://jpeelan.github.io/

EDUCATION

Purdue University West Lafayette, IN
Bachelor of Science in Electrical Engineering Expected: May 2028
Concentrations in Automatic Controls, Artificial Intelligence and Machine Learning
Minor in Mathematics

WORK EXPERIENCE

Swinergy Minneapolis, MN
Electrical Engineering Intern June 2026-August 2026

- Designed and prototyped a wireless ESP32-S3 sensor and actuator network for remote monitoring and control of a pilot-scale multi-stage digestion reactor
- Integrated industrial sensors, including 4–20 mA flow and level sensors, Type-K thermocouples, pH electrodes, and analog measurement circuits
- Developed a 24 V power and control architecture incorporating DC-DC conversion, circuit protection, MOSFET-driven relays, and pump and solenoid control

Purdue University College of Engineering West Lafayette, IN
Undergraduate Teaching Assistant January 2026-May 2026

- Guide students through hands-on electrical engineering labs involving circuit design, breadboard implementation, and troubleshooting using oscilloscopes, function generators, and power supplies
- Diagnose and resolve circuit faults by validating measurements, identifying wiring and design errors, and explaining underlying circuit behavior
- Provide clear, constructive technical feedback on lab work while reinforcing proper lab practices and measurement techniques
- Maintain professional communication with students and course staff while adapting instruction in real time

TECHNICAL EXPERIENCE

Purdue's Mechanisms and Robotic Systems ("MARS") Lab West Lafayette, IN
Undergraduate Researcher January 2026-Present

- Designed and prototyped a piezoelectric tactile sensing array using a row/column wiring matrix to reduce I/O requirements while enabling multi-point pressure detection across the sensor surface
- Developed a capacitive tactile sensing array and supporting sensing architecture for multi-point touch and force detection, including signal routing, PCB design, and experimental validation for robotic sensing applications.
- Designed and prototyped a flexible PCB-based piezoresistive tactile sensor for integration onto the LEAP Hand robotic platform, optimizing conformability, sensing coverage, and compact routing for articulated robotic fingers.

Purdue Humanoid Robotics Club West Lafayette, IN
Member August 2025-Present

- Collaborate with a multidisciplinary team to design and build a humanoid robot, focusing on developing precise electrical schematics for the robot's arms and legs to enable advanced articulation and movement

SKILLS

- Programming: Proficient in Python, C, and Matlab
- Circuit Design Software: Proficient in Altium, KiCad, and LTspice
- Hardware Description Languages: Proficient in Verilog for digital system design
- Electronics Testing & Measurement: Proficient with oscilloscopes, digital multimeters (DMMs), waveform/function generators, and bench power supplies
- Prototyping and Research: Proficient in soldering and prototyping circuits